Research Foundation for an EDM-2022 CLO Bloom's Taxonomy Benchmark

TL;DR

- **The dataset, source paper, and SOTA are pinned down.** The CSV at `/mnt/user-data/uploads/sample_full.csv` is from Li, Raković, Poh, Gašević & Chen (EDM 2022, pp. 530-537, DOI 10.5281/zenodo.6853191; Monash Centre for Learning Analytics; code at https://github.com/SteveLEEEEE/EDM2022CLO — **not** on Kaggle or HuggingFace). Their reported SOTA is binary BERT-base per Bloom level with F1 in **[0.878, 0.947]** and Cohen's κ up to **0.93**.
- **No prior work has ever benchmarked open-weight LLMs three ways on this dataset.** Almatrafi & Johri (2025) ran only GPT-4-turbo on a random 1,000-example subset; Kumar et al. (2025) ran four closed/open LLMs but on a different, small 600-sentence dataset using the original 1956 categories. A 12-model open-weight zero-shot/few-shot/QLoRA three-paradigm benchmark in both single-label and multi-label formulations is therefore a genuine first. nsf arxiv
- **All engineering parameters for the 12-hour single-A100 budget are determined and ready to drop into the notebook:** QLoRA NF4 + bf16 compute + double-quant; LoRA r=16, α=32, dropout 0.05, target_modules = q/k/v/o/gate/up/down_proj (with auto-detect fallback for Phi-3 fused QKV); lr 2e-4 cosine; paged_adamw_8bit; seq_len 256; stratified 1,500-example test subset; balanced 2-per-class few-shot exemplars; regex+JSON parser with parse-failure as a first-class metric.

Key Findings

1. **Dataset provenance is fully verified.** Yuheng Li et al., EDM 2022 short paper #55. 21,380 learning objectives scraped from 5,558 courses across Monash's 10 faculties (2021). One trained coder labeled all rows; a second coder labeled 30% (initial Cohen's $\kappa = 0.63$; after disagreement adjudication, 0.80). Multi-label cardinality: 2,325 LOs have 2 labels, 280 have 3, 2 have 4 (≈12% multi-label). github + 5
2. **The class distribution is heavily skewed.** Per Li et al. Table 2: Remember 886 / Understand 5,079 / Apply 5,074 / Analyze 2,311 / Evaluate 2,468 / Create 2,955. Remember is <5% of the data — this drives the choice of macro-F1 as primary metric and mandates class-aware stratification of the 1,500-example LLM test subset (with a Remember floor of ~100 examples). educationaldatamining
3. **Li et al.'s SOTA is BERT-base-uncased per-class binary classifiers** (Table 3): Remember F1 0.878 / Understand 0.947 / Apply 0.931 / Analyze 0.922 / Evaluate 0.929 / Create 0.888. SVM, Random Forest, and XGBoost trailed by only ~2–3 F1 points; Naive Bayes and Logistic Regression trailed by 20+ F1 points (NB F1 was as low as 0.20 on Remember). A multi-class multi-label BERT was slightly worse than the per-class binary BERTs on 5 of 6 levels. educationaldatamining + 3
4. **GPT-4 has been benchmarked on this exact dataset (Almatrafi & Johri 2025, Computers and Education: AI 8:100404)** on a random 1,000-example test subset. Best prompt was the combination of rhetorical-situation framing with 5-shot in-context examples (P2+P3): per-class Cohen's κ ranged **0.555–0.720**. Fine-tuned BERT-base on the same subset still won on every class (κ **0.849–0.921**). Chain-of-thought *hurt* GPT-4 on this task on five of six classes — a strong reason to keep CoT out of the main pipeline and tag it only as future work. ScienceDirect
5. **Open-weight LLMs in the small-data Bloom literature plateau around 0.70–0.75 zero-shot.** Kumar, Gulwani & Singh (2025, arXiv:2511.10903) on a different 600-sentence dataset using the original 1956 categories: best zero-shot accuracy was **GPT-4o-mini at 0.73, with Gemini-1.5-Pro at 0.72; Anthropic Claude-3.5-Haiku 0.58; LLaMA-3.1 (via Ollama) 0.42** — all dominated by an SVM with synonym augmentation at accuracy 0.94. Misclassifications "mostly happen at the adjacent cognitive levels," consistent with the Apply↔Analyze and Evaluate↔Create confusions noted in Li et al. arxiv
6. **PEFT/QLoRA defaults for May 2026 on A100** converge across Unsloth, HuggingFace TRL, Google AI for Developers, and the Dettmers et al. QLoRA paper: 4-bit NF4 weights, bf16 compute, double-quantization (Dettmers et al. 2023 §1: "Double Quantization, a method that quantizes the quantization constants, saving an average of about 0.37 bits per parameter (approximately 3 GB for a 65B model)"). LoRA r=16 / α=32 / dropout 0.05 is the modern default for ⩽9B sequence-classification fine-tuning; `target_modules="all-linear"` or the explicit Llama-style list q/k/v/o/gate/up/down_proj works for Qwen2.5, Llama 3, Gemma 2, Mistral, Phi-4-mini, SmolLM2 (Phi-3-mini's fused `qkv_proj`/ `gate_up_proj` requires runtime auto-detection). arxiv
7. **The notebook's contribution gap is unambiguous and defensible.** No prior study has (a) benchmarked ⩾10 open-weight instruct LLMs on the Monash dataset, (b) compared zero-shot vs few-shot vs QLoRA head-to-head on it, (c) evaluated both the single-label dominant-collapse and the full multi-label formulation, or (d) made the entire pipeline reproducible on a single A100 within 12 hours.

Details

A1. Authoritative reference for the dataset paper

Citation (BibTeX-ready): Li, Y., Raković, M., Poh, B. X., Gašević, D., & Chen, G. (2022). *Automatic Classification of Learning Objectives Based on Bloom's Taxonomy.* In Proceedings of the 15th International Conference on Educational Data Mining (EDM 2022), Durham, UK, pp. 530–537. DOI: 10.5281/zenodo.6853191.

- Corresponding authors: Mladen Raković and Guanliang Chen (Monash Centre for Learning Analytics).
- Code & data: GitHub `SteveLEEEEE/EDM2022CLO`, file `data/sample_full.csv`. Not mirrored on Kaggle or HuggingFace (search confirmed).
- Train/test protocol: single 80:20 random split, reused across all models. For BERT, the 80% was further split 80/20 to obtain a validation set. Two formulations evaluated: per-level binary classifiers (5 ML + 1 BERT = 36 binary classifiers) and a single multi-class multi-label BERT (sigmoid + 0.50 threshold). educationaldatamining + 2
- Feature stack for ML (3,094 features): 1,000 unigrams + 1,000 bigrams + 1,000 TF-IDF + Automated Readability Index + 93 LIWC-2015 features. educationaldatamining
- BERT setup: bert-base-uncased, 12L × 768, dropout 0.1, batch 64, 3 epochs, early stopping on F1 with patience 10. educationaldatamining

A2. Methodology gaps in Li et al. (2022)

1. No LLMs (predates the instruct era).
2. No single-label "dominant level" collapse experiment.
3. Single primary coder for ~70% of the data (single-coder bias acknowledged).
4. Per-class multi-class multi-label results not reported as a single macro-F1.
5. No class-imbalance mitigation beyond per-level binary decomposition.
6. No comparison to modern open-weight small LMs (which now compete with BERT-base on many text tasks).

A3. Bloom prior-literature numbers (for the notebook's Related-Work markdown)

Year	Authors	Venue	Dataset	Best model	Best result
2020	Mohammed & Omar	PLOS ONE 15(3):e0230442	600 exam Qs	SVM + TFPOS-IDF + Word2Vec	weighted-F1 ≈ 0.897
2021	Shaikh, Daudpotta, Imran	IEEE Access 9:117887	CLOs + question items	LSTM + Wiki word vectors	weighted-F1 0.73 (CLOs); macro-F1 0.82 (questions); overall acc 0.87/0.74 Academia.edu
2021	Waheed et al. ("BloomNet")	ICNLSP 2021 / arXiv:2108.07249	Two CLO datasets	Transformer + linguistic features (POS+NER+WA)	weighted-F1 0.71 / 0.85 across two datasets
2021	Zhang et al. ("BT-BERT")	ACE'21	Computing-education Qs	BERT	82.6% acc on frequent levels, 59.2% all-level educationaldatamining
2022	**Li et al.**	**EDM 2022**	**21,380 Monash CLOs**	**Binary BERT-base**	**F1 0.878–0.947, κ up to 0.93**
2023	(IndoBERT FT)	local Indonesian journal	Indonesian exam Qs	IndoBERT FT	acc/F1 ≈ 0.97 on small balanced set ResearchGate
2025	Patil et al. follow-up	Electronics (MDPI 14:2312)	755 short answers	KNN+SVM+NB ensemble + TFPOS-IDF	acc 0.935 MDPI
2025	**Almatrafi & Johri**	Computers and Education: AI 8:100404	1,000-subset of Monash	GPT-4-turbo + rhetorical situation + 5-shot (P2+P3)	per-class κ 0.555–0.720; fine-tuned BERT still beat it (κ > 0.84 all classes)
2025	Kumar, Gulwani & Singh	arXiv:2511.10903 (UNSW)	600 LOs/ exam Qs (original 1956 labels)	SVM + synonym aug	acc 0.94, macro-F1 0.94; best LLM GPT-4o-mini acc 0.73

Almatrafi & Johri 2025 per-class numbers (Cohen's κ) — most directly comparable prior numbers:

Prompt	Remember	Understand	Apply	Analyze	Evaluate	Create
P1 zero-shot	0.484	0.685	0.642	0.531	0.558	0.540
P2+P3 best	**0.573**	**0.720**	0.661	0.583	0.556	0.555
P4 CoT (hurts)	0.347	0.599	0.620	0.421	0.553	0.582
Fine-tuned BERT	0.917	0.896	0.888	0.876	0.921	0.849

Important cross-paper caveat: Almatrafi & Johri (2025) and Li et al. (2022) use the **revised 2001** taxonomy (Remember/Understand/Apply/Analyze/Evaluate/Create) — same as the benchmark — but Kumar et al. (2025) use the **original 1956** categories (Knowledge/Comprehension/Application/Analysis/Synthesis/Evaluation). Do not directly equate. arxiv

A4. PEFT / QLoRA configuration matrix (drop-in for notebook)

BitsAndBytes (canonical for all six ≤4B targets):

```python
BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16,
    bnb_4bit_use_double_quant=True,
)
```

LoRA hyperparameters (consensus from Unsloth, TRL, HF PEFT, Hu et al. 2022, Dettmers et al. 2023):

Hyperparameter	Value	Source
r	16	Unsloth defaults; Hu et al. 2022 ICLR (arXiv:2106.09685)
alpha	32	2r convention, empirically robust
dropout	0.05	TRL/HF guides
bias	"none"	standard
target_modules	q,k,v,o,gate,up,down_proj	Llama family; auto-detect for Phi-3 fused QKV
task_type	"CAUSAL_LM"	SFT chat-formatted JSON answer route
lr	2e-4	QLoRA paper / Unsloth
optim	paged_adamw_8bit	QLoRA paper
epochs	1–2 on 5,000-sample train subsample	budget
schedule	cosine, warmup ratio 0.03	TRL
bf16	True	A100; never fp16 with NF4
grad_checkpoint	True	memory
seq_len	256	dataset 99th-pct ≈ 55 tokens

Per-model batch sizes on a single A100 40GB (effective batch 16 via grad accum):

Model	per_device_train_batch	grad_accum	Notes
smollm2-1.7b-instruct	8–16	1–2	Easy fit; can fit non-QLoRA
qwen25-1.5b-instruct	8–16	1–2	Easy fit
qwen25-3b-instruct	4–8	2–4	Comfortable
gemma2-2b-instruct	4–8	2–4	Comfortable
phi35-mini-4k-instruct (3.8B)	4	4	Note fused qkv_proj/ gate_up_proj
phi4-mini-instruct (≈3.8B)	4	4	Same family as Phi-3
mistral-7b-instruct-v0.3	2	8	grad_ckpt on
qwen2-7b / qwen25-7b	2	8	grad_ckpt on
llama3-8b-instruct (gated)	2	8	grad_ckpt on
gemma2-9b-instruct (gated)	1	16	Tight on 40GB; flag-gated
mistral-nemo-instruct (≈12B)	1	16	Tight on 40GB; flag-gated

Gemma 2 9B and Mistral-Nemo each consume ≈2–3 h of the 12-h budget when QLoRA-fine-tuned and so should be gated behind `RUN_LARGE_QLORA=False` by default. On A100 80GB, double the per_device batch and halve grad_accum.

A5. Prompting / structured-output recipe

Zero-shot system prompt skeleton (single-label):

> You are an expert in educational psychology and Bloom's revised taxonomy (Anderson & Krathwohl, 2001). Classify the given course learning outcome into exactly one of: Remember, Understand, Apply, Analyze, Evaluate, Create.
> Definitions:
>
> - **Remember**: Retrieve relevant knowledge from long-term memory (recognize, recall). arxiv arXiv
> - **Understand**: Construct meaning (interpret, exemplify, classify, summarize, infer, compare, explain). arxiv
> - **Apply**: Carry out or use a procedure in a given situation (execute, implement). arxiv arXiv
> - **Analyze**: Break material into parts and detect how parts relate to one another and to an overall structure (differentiate, organize, attribute). arxiv
> - **Evaluate**: Make judgments based on criteria and standards (check, critique). arxiv arXiv
> - **Create**: Put elements together to form a coherent or functional whole; reorganize into a new pattern (generate, plan, produce). arxiv
>
> If ambiguous between levels, pick the **highest** level whose cognitive demand is fully expressed by the main verb and its object.
> Output ONLY valid JSON: {"level": "<one of the 6>"}. No reasoning, no prose, no markdown.

Multi-label variant swaps `"level"` for `"levels": [...]` and instructs the model to include every level whose cognitive demand the outcome expresses.

Decoding: `apply_chat_template(..., add_generation_prompt=True)`, `do_sample=False`, `max_new_tokens=64` (single-label) / `96` (multi-label), `torch_dtype=torch.bfloat16`, `device_map="auto"`.

Parser: Two-stage — try `json.loads` first; on failure regex-search `(?i)\b(Remember|Understand|Apply|Analyze|Evaluate|Create)\b` and take the first hit; on full failure record `parse_fail=True` and assign a uniform random level. Log parse-failure rate per model as a first-class metric.

Few-shot exemplars: Balanced 2-per-class (12 total) sampled once with `seed=42` from the train split, persisted to `splits/fewshot_exemplars.json`. Shuffle the 12 once with a fixed seed; do **not** sort by class or length. Lu, Bartolo, Moore, Riedel & Stenetorp (ACL 2022, arXiv:2104.08786) showed example order can swing performance by 13% relative across eleven classification tasks; Zhao, Wallace, Feng, Klein & Singh (ICML 2021, arXiv:2102.09690) showed that "varying the permutation of the training examples can cause accuracy to go from near chance (54.3%) to near state-of-the-art (93.4%)" on 4-shot SST-2 with GPT-3 2.7B and identified three biases (majority label, recency, common token) that motivate balanced order. Apply the same 12-exemplar block to every LLM for fair head-to-head comparison. arXiv + 2

Label-length bias: Sun et al. 2025 (arXiv:2511.14385) — Bloom labels all tokenize to 1–2 BPE tokens, so the bias is small here, but it is worth measuring per-class F1 deltas as an ablation if the leaderboard shows asymmetric per-class deficits.

A6. Evaluation metric checklist

Single-label (primary):

- Accuracy
- **Macro-F1 (primary headline)** — equal weight per Bloom level, robust to skew
- Weighted-F1 — matches Li et al. 2022 / Mohammed & Omar 2020 reporting style
- Cohen's κ — direct comparability with Li et al. and Almatrafi & Johri
- Per-class precision/recall/F1 + confusion matrix CSV

Multi-label (secondary):

- Subset accuracy (exact match)
- Micro-F1 — globally pooled, primary multi-label metric
- Macro-F1 — equal weight per label
- "Samples" F1 (per-instance F1 averaged)
- Hamming loss
- Jaccard (sample-averaged)

Wu & Zhu (arXiv:2011.07805): when label cardinality is low (mean ≈1.13 here), Hamming loss and subset accuracy do not meaningfully conflict — report both.

A7. Stratification & tie-break

- `bloom_single` **derivation:** for each row with ⩾1 positive Bloom label, take the **highest** level in Create > Evaluate > Analyze > Apply > Understand > Remember. This is the Anderson & Krathwohl (2001) cognitive-complexity hierarchy; higher levels are cognitive supersets of lower ones in the revised taxonomy.
- **Splits:** 70/10/20 stratified on `bloom_single` (vs Li et al.'s 80/20 no-val) — yields a proper validation split for early stopping and prompt selection without leaking the test set. Persist `splits/{train,val,test}_idx.npy`.
- **LLM test subset:** 1,500 examples stratified on `bloom_single` from the 20% test split, with a **Remember floor of 100** examples for stable per-class F1. Persist as `splits/llm_test_idx.npy`. Identical subset used for every LLM and every paradigm.
- **Few-shot exemplars:** balanced 2 per class from train, persisted to `splits/fewshot_exemplars.json`.

A8. Contribution framing (drop-in for Section 0 Abstract & Contributions)

Aspect	**Li et al. 2022**	**Almatrafi & Johri 2025**	**Kumar et al. 2025**	**This benchmark**
Dataset	21,380 Monash CLOs	1,000-subset Monash	600 sentences (1956 cats)	21,380 Monash CLOs
Classical ML	5 models	—	3 models w/ aug	3 models (LR, SVM, RF) on full data
BERT-family	bert-base	bert-base	BERT + RoBERTa	DistilBERT
LLM zero-shot	—	GPT-4 (closed)	4 LLMs (3 closed, 1 open)	**12 open instruct LLMs**
LLM few-shot	—	GPT-4 only	—	**12 open instruct LLMs**
LLM QLoRA	—	—	—	**6 open LLMs ⩽4B**
Single-label	—	per-class binary	yes	**yes (dominant-level collapse)**
Multi-label	yes	per-class binary	—	**yes (sigmoid + JSON multi-hot)**
Compute transparency	no	no	no	**single A100, ⩽12 h, resumable**

A9. Engineering checklist (for Section 1 of the notebook)

1. **Pinned versions (May 2026):** `torch>=2.4`, `transformers>=4.46`, `peft>=0.13`, `bitsandbytes>=0.43`, `accelerate>=1.0`, `trl>=0.11`, `datasets>=3.0`, `scikit-learn>=1.5`, `sentencepiece>=0.2`, `protobuf>=4.25`.
2. **HF auth:** `os.environ.get("HF_TOKEN")` then `huggingface_hub.login(token=…)` else `notebook_login()`. Document gated models: Llama-3-8B-Instruct, Gemma-2-2B/9B-Instruct, Mistral-7B-Instruct-v0.3, Mistral-Nemo-12B-Instruct require accepting the model card license before download.
3. **Seed function** seeds `random`, `numpy`, `torch`, `torch.cuda`, `transformers.set_seed(42)`, and sets deterministic cuBLAS flags. Note: bf16 matmul on A100 is **not bit-deterministic** — document expected metric variance <0.5 F1 across reruns.
4. **OOM defense:** wrap every model load and every `.generate()` / `Trainer.train()` in try/except for `torch.cuda.OutOfMemoryError` and generic exceptions; on failure, write a row with NaN metrics + `status="OOM"` (or `"ERROR: <msg>"`), then `model.cpu(); del model; del tokenizer; gc.collect(); torch.cuda.empty_cache(); torch.cuda.synchronize()` and continue with the next model.
5. `is_already_evaluated` tests for non-empty metrics CSV AND `status == "success"`; partial/failed runs are re-run.
6. **Confusion-matrix CSV format:** rows = true Bloom label, columns = predicted label, fixed Bloom-order index.
7. **Per-model wall-clock & peak-VRAM logging** alongside metrics for compute-aware reporting.

A10. Hypotheses for the discussion section

Grounded in Almatrafi & Johri 2025 and Li et al. 2022:

1. Fine-tuned DistilBERT will beat zero-shot LLMs on macro-F1 by 8–15 points.
2. Few-shot will beat zero-shot for ⩾10 of 12 LLMs; the gain will be largest for the smallest models (Qwen2.5-1.5B, SmolLM2-1.7B).
3. QLoRA-fine-tuned small LLMs (⩽4B) will approach DistilBERT within 1–3 macro-F1 points.
4. Apply↔Analyze and Evaluate↔Create remain the dominant confusion pairs across every paradigm.
5. Remember will show highest accuracy but lowest κ across LLM prompting (rare class + obvious verbs).
6. Classical SVM with TF-IDF will land in macro-F1 0.75–0.82, weighted-F1 ≈ 0.85, below DistilBERT but above zero-shot LLMs.

Recommendations

Build the notebook in this order, with these specific defaults:

1. **Section 0–2 first (Markdown + utilities).** Pin the Li et al. 2022 citation, BloomNet (Waheed et al. ICNLSP 2021), Mohammed & Omar (2020), Anderson & Krathwohl (2001), Bloom (1956), Almatrafi & Johri (2025), Kumar et al. (2025), QLoRA (Dettmers et al. 2023, arXiv:2305.14314), LoRA (Hu et al. 2022, arXiv:2106.09685). Implement `save_results`, `is_already_evaluated`, `aggregate_all_results` exactly per the spec — these unlock resumability.
2. **Section 3 EDA.** Build `bloom_single` with the Create > Evaluate > … > Remember tie-break. 70/10/20 stratified split on `bloom_single`. Persist split indices. EDA: class bar plot, co-occurrence heatmap (multi-label), token-length histogram with `bert-base-uncased`, top distinctive verbs per class via log-odds-ratio with informative Dirichlet prior (Monroe-Colaresi style) or simpler TF-IDF top-k.
3. **Section 4 classical ML.** TF-IDF(1,2)-grams, sublinear TF, min_df=2, max_df=0.95. Single-label: LR (saga, L2), LinearSVC + CalibratedClassifierCV, RandomForest. Multi-label: same backbones wrapped in `OneVsRestClassifier`. Save all CSVs via `save_results`.
4. **Section 5 DistilBERT.** Two configs (single-label CE, multi-label BCE) with `bf16=True`, `per_device_batch=32`, 3–4 epochs, `metric_for_best_model="f1_macro"`, early stopping patience 2. Threshold 0.5 for multi-label.
5. **Section 6 LLM framework.** Build the 1,500-example stratified subset (Remember floor 100); build the 12-exemplar fixed few-shot block. Iterate the `MODEL_REGISTRY` for zero-shot first (fast, validates pipeline), then few-shot. QLoRA last on the six ⩽4B models, gated `RUN_LARGE_QLORA=False` for the two ⩾9B candidates.
6. **Section 7–8 leaderboard + discussion.** Pivot table: rows = model, columns = paradigm × formulation, cells = macro-F1. Highlight the three best-confusion-matrices. In the discussion, explicitly state which paradigm wins per model-size bucket; cite Almatrafi & Johri (2025) numbers in-line as the GPT-4-closed-model anchor against which the open-weight 12-model open-source landscape is measured.

Benchmarks that should change the plan if they fail:

- If DistilBERT macro-F1 < 0.75 → re-check label cleaning (stripping leading quotes / whitespace) and verify the `bloom_single` tie-break is actually being applied.
- If zero-shot parse-failure rate > 15% for any model → switch that model's pipeline to Outlines constrained decoding before drawing conclusions about its capability.
- If QLoRA macro-F1 < zero-shot for any small model → the SFT loss is almost certainly mis-targeted (training on the prompt tokens, not only the JSON answer). Use `DataCollatorForCompletionOnlyLM` (TRL) to mask the prompt.
- If the 12-hour budget is exceeded by Section 6.3 (QLoRA) → halve `MAX_LLM_TRAIN_SAMPLES` from 5,000 to 2,500 and/or drop epochs from 2 to 1; the QLoRA literature shows 1-epoch fine-tunes capture most of the gain.

Caveats

1. **The Monash dataset is the work of one university and one primary coder for ~70% of labels.** External validity to other institutions, other disciplines, or assessment items (vs. learning outcomes) is unverified; do not over-claim transferability in the conclusion.
2. **Bf16 matmul on A100 is not bit-deterministic**; expected metric variance across reruns is $<$ 0.5 F1 but exact replication is not guaranteed. Document the seed and the empirical variance from at least one rerun if budget allows.
3. **The 1,500-example LLM test subset trades statistical power for compute feasibility.** Per-class F1 confidence intervals are ±2–3 F1 points; do not over-interpret small inter-model gaps. The aggregate macro-F1 is reliable.
4. **GPT-4 results from Almatrafi & Johri (2025)** were measured on a *different* random 1,000-example subset of the same dataset, not the stratified 1,500 of this benchmark. Comparisons to those numbers are indicative, not strictly head-to-head; reproduce GPT-4 on the same 1,500 subset before formally tabulating side-by-side.
5. **Kumar et al. (2025) numbers use the original 1956 Bloom categories** (Knowledge / Comprehension / Application / Analysis / Synthesis / Evaluation), not the revised 2001 categories used here. Do not align their per-class numbers with ours.
6. **Single-coder labeling noise upper-bounds any classifier's headline accuracy.** Li et al.'s post-adjudication Cohen's κ between coders was 0.80 on a 30% subsample; classifier κ above ~0.85 is approaching the noise ceiling and should not be over-interpreted. nsf educationaldatamining
7. **CoT prompting and self-consistency are deliberately out of scope** for the main runs because Almatrafi & Johri 2025 showed CoT degrades GPT-4 on five of six classes for this exact task. Flag them as Section 8 future work. nsf
8. **Kumar, Gulwani & Singh (2025) is an arXiv preprint (v1, Nov 14 2025)** and has not been peer-reviewed; treat its numbers as indicative.
9. **Mistral-Nemo and Gemma 2 9B QLoRA each consume ≈2–3 h of the budget on A100 40GB** and are flag-gated. Without an 80GB A100, the headline LLM leaderboard should be presented over the 10 ⩽8B models and the two ⩾9B models reported separately if the user opts to enable them.
10. **The dataset CSV is GitHub-only**; if this notebook is published, mirror `sample_full.csv` to a stable archive (Zenodo or HuggingFace dataset) with the original Li et al. citation before publication, to ensure long-term reproducibility.